

NeuroCore – Formal Mathematical Framework for Transition-Sensitive Regime Inference

Abstract

NeuroCore is a domain-agnostic computational framework for detecting functional state transitions in high-entropy time-series by modeling structural stability over time.

1. Problem Formulation

Let $x(t) \in \mathbb{R}^d$ be a stochastic time-series and $\lambda(t)$ latent regimes. Objective: detect transitions and stability.

2. Functional Representation

$x(t) \rightarrow z(t)$, where $z(t)$ is a feature embedding and $\lambda(t)=f(z(t))$.

3. Core Operators

$\Phi(t)$: $\sum \log(1 + |\nabla x(i)|) \rightarrow$ local structural variation.

$\Delta(t)$: $|\Phi(t)-\Phi(t-1)| \rightarrow$ transition sensitivity.

$K(t)$: $\sigma/\mu \rightarrow$ global dispersion.

$L(x_t) \rightarrow \phi \approx 0.55 \rightarrow$ asymptotic stability attractor.

4. Feature Fusion

$z(t) = [\Delta, K, L, z_E]$

5. Regime Definition

Regimes are metastable regions in feature space.

6. Transition Sensitivity

Small variations produce measurable structural changes.

7. Theoretical Positioning

Links to dynamical systems, stochastic processes, and nonlinear time-series analysis.

8. Conclusion

NeuroCore models signals as evolving functional structures rather than static values.